

LITTLE STREAM DETECTOR**LSD 3.0****Detailed Technical Documentation**

Release	3.0 - VP9 MP4 mapping build
Reference file	LSD-3.0-VP9-MP4.ps1
Build SHA-256	bf7955bd859a2f37768c981d96d6ab39e23285d7ce334bae7ce3b98d894e62ff
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Runtime	Windows PowerShell 5.1 and .NET Framework
External multimedia tools	None
Temporary media files	None

Scope. This revision documents native VP9 analysis for Matroska/WebM and the newly implemented ISO Base Media sample-entry mapping for VP9 in non-fragmented MP4/M4V/MOV.

Validation state. Matroska/WebM VP9 remains regression validated. MP4 VP9 mapping is implemented for vp09 and legacy vp08 sample entries, but runtime validation requires a representative VP9-in-MP4 asset.

Contents

- 1. Purpose and System Scope
- 2. Architectural Overview
- 3. Runtime and Processing Lifecycle
- 4. Canonical Media Model
- 5. Container Modules
- 6. Video Analysis Modules
- 7. Audio Analysis Modules
- 8. Quantizer and Bitrate Analysis
- 9. GUI, Jobs, Reporting, and Comparison
- 10. Reference Comparison State Model
- 11. Validation and Defensive Parsing
- 12. Supported Combinations
- 13. Known Limitations
- 14. Verification and Regression Matrix
- 15. Maintenance Guidelines
- Appendix A. Key Types and Responsibilities
- Appendix B. Terminology
- Appendix C. Release Identification

Updated in this revision. VP9 is now a supported native video family for Matroska/WebM. Documentation now covers VP9 superframes, hidden coded frames, show-existing events, KEY/INTER classification, reference-slot geometry, Base Q Index extraction, completeness checks, and the validated regression assets.

1. Purpose and System Scope

Little Stream Detector 3.0 is a native Windows PowerShell application with an embedded C# analysis engine. It inspects media containers, builds a container-neutral sample index, performs codec-aware bitstream analysis, and presents human-readable and structured results through a responsive graphical interface.

The VP9 extension adds deterministic traversal of VP9 uncompressed frame headers without changing container addressing, the canonical model, or A/B comparison semantics.

1.1 Primary objectives

- Provide deterministic native analysis without FFmpeg, MediaInfo, or other external multimedia processes.
- Keep container-specific addressing separate from codec-specific interpretation.
- Validate every canonical sample against file boundaries before reading payload data.
- Report unavailable information as N/A rather than guessing.
- Support asynchronous preparation and analysis without blocking the GUI.
- Retain Windows PowerShell 5.1 and in-memory EXE hosting compatibility.
- Allow visual comparison of exactly one Current A and one Reference B using completed native results.
- For VP9, distinguish container samples, internal coded frames, shown frame events, and hidden frames.

1.2 Supported families

Area	Supported families
Containers	Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV
Video	H.264/AVC, H.265/HEVC, AV1, VP9, MPEG-4 Part 2/XviD
Audio	AAC-LC, HE-AAC, HE-AACv2, MP3, PCM integer/float, AC-3, E-AC-3
Outputs	Summary A, optional Summary B, Streams, JSON, Log, bitrate profile, quantizer distribution

2. Architectural Overview

LSD remains a modular monolith. PowerShell owns application state, presentation, orchestration, and background-job lifecycle. Embedded C# types own binary parsing, canonical models, bitstream traversal, validation, and numerical accumulation.

Pipeline. User/GUI -> metadata and job coordinator -> Matroska, MP4, or AVI reader -> canonical media model -> video and audio analyzers -> completed result snapshot -> Current A / Reference B comparison views.

2.1 Logical layers

Layer	Responsibility	Representative components
Presentation	Window, reports, graphs, drag-and-drop, active-tab copy	PowerShell WinForms
Coordination	Metadata preparation, state transitions, cancellation, progress	Start-Meta, Complete-NativeMetadata, Start-Scan
Container parsing	Tracks and canonical sample locations	NativeMediaProbe, NativeMp4Probe, NativeAviProbe
Canonical adaptation	Normalize tracks and bounded samples	MatroskaToLsdAdapter, Mp4ToLsdAdapter, AviToLsdAdapter
Codec analysis	Configuration and frame-level syntax	AVC, HEVC, AV1, VP9, MPEG-4 Visual
Comparison	Retain B and render A/B views	ReferenceResult, ReferenceOnlyView

2.2 VP9 isolation

The V_VP9 mapping already produces CodecFamily VP9 and PayloadFormat Vp9Frame. The new Vp9FrameParser is selected only when the active video track is VP9. Other codec branches and all container readers remain unchanged.

3. Runtime and Processing Lifecycle

3.1 Startup

- Compile embedded C# through Add-Type.
- Run native self-tests for selected readers and arithmetic behavior.
- Hide the console window for GUI operation.
- Apply invariant English culture for deterministic formatting.
- Initialize current-result, reference-result, VP9 diagnostics, jobs, graph, and report state.

3.2 File processing phases

Phase	Action	Output
1. Selection	Select or drop a file	Absolute source path
2. Native detection	Identify container	Native MediaInfo
3. Metadata preparation	Parse tracks and sample indexes	Container probe result
4. Canonical adaptation	Create common bounded samples	LsdMediaModel
5. Codec analysis	Parse frame and audio syntax	Diagnostics and statistics
6. Finalization	Build report and histograms	Current A result
7. Optional promotion	Set Ref B	Reference B snapshot
8. Comparison	Analyze later A	A/B views

3.3 VP9 sample lifecycle

Each bounded Vp9Frame sample is read once. A valid superframe index may split that sample into multiple internal VP9 frames. Each internal frame is parsed in coded order so reference-slot geometry remains available. The sample contributes one bitrate observation, while every newly coded internal frame contributes one Base Q Index observation.

4. Canonical Media Model

4.1 LsdMediaModel

Field	Meaning
SourcePath	Input file path
ContainerType / ContainerDetail	Normalized family and specific route
FileLength	Source length in bytes
DurationSeconds	Best available duration
TimecodeScale	Nanoseconds per canonical timestamp tick
Tracks	Canonical video, audio, subtitle, and other tracks

4.2 LsdTrackModel and LsdSampleModel

Object	Key fields
LsdTrackModel	TrackId, TrackType, CodecId, CodecFamily, CodecPrivate, PayloadFormat, NaLengthSize, PCM parameters, Samples
LsdSampleModel	FileOffset, Length, DecodeTimestamp, PresentationTimestamp, Duration, key/invisible/discardable flags, PayloadFormat

4.3 VP9 accounting semantics

A canonical VP9 sample is a container payload range, not necessarily one coded frame. Superframes can carry one shown frame and one or more hidden coded frames. LSD therefore reports Video samples, Internal VP9 frames, Shown frame events, Hidden coded frames, and Show existing frame separately.

4.4 Reference result snapshot

Reference B remains a presentation-layer snapshot containing file identity, duration, bitrate bins, quantizer mode, quantizer counts, and frozen Summary B text. VP9 uses QpMode "VP9 Base Q Index".

5. Container Modules

5.1 Matroska and WebM

The Matroska reader maps V_VP9 to CodecFamily VP9 and PayloadFormat Vp9Frame. This route remains regression validated.

5.2 MP4, M4V, and MOV

The ISO Base Media reader supports non-fragmented files with a moov-based sample table. The video sample-entry reader now recognizes vp09 and legacy vp08, maps them to V_VP9, reads the optional vpcC child box, and routes bounded samples to the existing Vp9Frame parser. No second VP9 parser path is introduced.

The existing stsd, stsz, stsc, stco/co64, stts, ctts, and stss logic continues to resolve sample locations and timing. The optional vpcC record is retained as codec private data; VP9 frame analysis uses the coded sample payloads.

5.3 AVI 1.0 and OpenDML AVI 2.0

VP9 in AVI is not enabled or validated.

5.4 Comparison neutrality

Completed VP9 results use the same Current A / Reference B snapshots regardless of whether the canonical samples originated in Matroska/WebM or non-fragmented MP4.

6. Video Analysis Modules

Family	Native analysis
H.264 / AVC	avcC and Annex B, SP S/PPS, I/P/B, SliceQP, CABAC/CAVLC, VUI
H.265 / HEVC	hvcC, VPS/SPS/PPS, NAL class, first-slice I/P/B, SliceQP, VUI
AV1	av1C, sequence/frame OBU traversal, hidden/show-existing, Base Q Index
VP9	Superframe index, uncompressed header, reference slots, shown/hidden events, KEY/INTER, Base Q Index
MPEG-4 Part 2 / XviD	VOI/VOP, I/P/B/non-coded VOPs, packed samples, VOP quantizer

6.1 VP9 parser scope

- Validate the two-bit frame marker and profile signaling.
- Recognize show_existing_frame without inventing a new quantizer value.
- Parse KEY_FRAME, INTER_FRAME, and intra-only syntax needed to reach quantization.
- Track eight reference slots so frame_size_from_refs can resolve width and height.
- Split valid superframes using the trailing duplicated marker and little-endian frame sizes.
- Read base_q_idx and the three frame-level delta-Q fields.
- Classify displayed KEY frames as I and displayed INTER frames as P. VP9 has no classic B-frame class in this report model.

6.2 Quantizer meaning

VP9 Base Q Index is a frame-level lookup-table index from 0 through 255. Segmentation may override quantization for individual blocks, so LSD does not label this histogram as a block-level quantizer distribution.

7. Audio Analysis Modules

Family	Validation and reporting
AAC	AudioSpecificConfig, core/output rate and channels, SBR, Parametric Stereo, profile naming
MP3	Version/layer, bitrate and sample-rate index, padding, frame size, decoded samples
AC-3 / E-AC-3	Syncword, bitstream ID, sample rate, frame size, blocks, channels, LFE
PCM	Block alignment, payload divisibility, decoded duration, expected and measured bitrate

Audio analysis remains independent of the VP9 extension. Audio details for Reference B remain available through Summary B. No waveform or quality score is synthesized.

Regression observation. The full VP9 reference file retained AAC-LC accounting with 27,247 canonical samples, zero rejected samples, and a measured bitrate of 128.000 kbps.

8. Quantizer and Bitrate Analysis

8.1 Bitrate profile

Each canonical video sample contributes byte size to presentation-time bins. Internal hidden VP9 frames remain inside their containing sample and do not create synthetic timing or bitrate bins. Reference B is gold and Current A is blue.

8.2 Quantizer sources

Codec	Native value	Histogram unit
AVC	SliceQP	Frame-level SliceQP
HEVC	SliceQP	Frame-level SliceQP
AV1	Base Q Index	New coded frame
VP9	Base Q Index	New internal coded frame
MPEG-4 Part 2	VOP quantizer	Coded VOP

8.3 VP9 completeness

- A Base Q Index is emitted only for a newly coded VP9 frame.
- `show_existing_frame` produces a display event but no new Base Q Index.
- A superframe can emit multiple Base Q Index values from one canonical sample.
- The histogram is enabled only when rejected headers equal zero and at least one internal coded frame was parsed.
- Frame accounting requires $\text{InternalFrames} = \text{Parsed} + \text{ShowExisting}$.
- Shown/sample consistency requires $\text{ShownFrameEvents} = \text{VideoSamples}$ for the validated Matroska route.

8.4 Dual DRF/QP distribution

The right panel uses the current quantizer mode in its title. For VP9 it displays "VP9 Base Q Index distribution". The contextual range is clamped to the valid codec domain 0 through 255.

9. GUI, Jobs, Reporting, and Comparison

9.1 Views

View	Content
Summary A	Readable report for Current A, including VP9 diagnostics
Summary B	Frozen reference report while B exists
Streams	Structured current native-track data
JSON	Serializable current metadata and streams
Log	Current lifecycle and validation messages
Bitrate graph	Blue A and gold B with role legend
Quantizer panel	Mode-aware A/B statistics, bars, and percentages

9.2 VP9 report block

The Native VP9 Frame / Base Q Analysis block reports samples, internal frames, superframes, parsed and rejected headers, show-existing events, shown and hidden frames, KEY/INTER counts, Base Q statistics, first failure, and four explicit validation statuses.

9.3 Copy action

Copy follows the active report tab. VP9 adds no alternate copy path and does not alter Summary B freeze behavior.

10. Reference Comparison State Model

The workflow supports exactly one Current A and one Reference B. It is not a queue and does not rank files.

State sequence. EMPTY -> open -> A blue -> Set Ref B -> B gold -> open -> B gold + A blue -> Remove Ref B -> A blue, or EMPTY when B was the only result.

10.1 State rules

- Every newly opened or dropped file becomes Current A and is blue.
- Set Ref B promotes completed A to B, changes the role color to gold, and waits for a new A.
- Opening another file while B exists preserves B and creates a new blue A.
- Remove Ref B removes only B.
- B is never silently converted back into A.

10.2 VP9 compatibility

Two VP9 results are directly comparable when both snapshots carry QpMode "VP9 Base Q Index". AV1 Base Q Index and VP9 Base Q Index remain distinct modes even though both use a 0 through 255 index domain. LSD does not force a cross-codec overlay.

11. Validation and Defensive Parsing

11.1 Boundary checks

- Canonical samples require non-negative offsets and positive lengths.
- Offset plus length must remain within the source file.
- VP9 sample lengths must fit the in-memory parser input contract.
- A superframe index must fit inside the sample, repeat the trailing marker, contain positive frame sizes, and account exactly for the payload before the index.
- Every internal frame reader is bounded to its indexed frame length.
- Frame marker, reserved profile bits, frame sync code, RGB profile constraints, reference-slot availability, and Base Q range are validated.

11.2 Error model

A VP9 syntax failure increments Headers rejected, preserves the first failure text, marks the sample frame type as Other, and prevents histogram activation. No guessed quantizer value is substituted.

11.3 State ordering

Internal frames are parsed in coded order. Reference slots are updated only from successfully parsed newly coded frames using `refresh_frame_flags`. This is required for `frame_size_from_refs` and for show-existing classification when a referenced slot has known state.

11.4 Runtime isolation

- No external multimedia executable is launched.
- No temporary media file is created.
- The source is opened read-only with shared read access.
- The single-file script remains compatible with in-memory EXE hosting.

12. Supported Combinations

12.1 Container and video matrix

Container	AVC	HEVC	Other
H.264 / AVC	Yes	Yes	Yes, Annex B
H.265 / HEVC	Yes	Yes	No
AV1	Yes	Yes	No
VP9	Yes, validated	Mapped; runtime test pending	No
MPEG-4 Part 2 / XviD	Yes	Yes	Yes

12.2 VP9 ISO Base Media route

- Recognized sample entries: vp09 and legacy vp08.
- Native codec identity: V_VP9.
- Canonical family and payload: VP9 / Vp9Frame.
- Optional configuration record: vpcC.
- Supported container structure: non-fragmented moov-based sample tables.
- Fragmented MP4 using moof/traf/trun remains unsupported.

12.3 Comparison combinations

Quantizer comparison still requires exactly matching QpMode. VP9 uses VP9 Base Q Index and is not overlaid with AV1 Base Q Index.

13. Known Limitations

When evaluating LSD 3.0, it is necessary to identify the areas in which broad multimedia frameworks such as FFmpeg or MediaInfo provide wider format coverage, greater container compatibility, or more suitable automation workflows.

- Fragmented MP4: moof, traf, tfhd, tfdt, and trun are not supported. MP4/MOV support is limited to non-fragmented files with conventional sample tables in moov.
- VP9 validation scope: Matroska/WebM V_VP9 remains regression validated for Profile 0, 8-bit YUV 4:2:0. MP4 vp09/vp08 mapping is implemented, but practical runtime validation is pending a representative VP9-in-MP4 test asset. VP9 in AVI is not enabled.
- VP9 quantizer meaning: the histogram is frame-level Base Q Index, not a reconstructed per-block distribution after segmentation.
- One comparison pair only: the UI supports exactly one Current A and one Reference B. It has no batch queue, directory-wide ranking, or simultaneous multi-file comparison.
- No pixel decoding: LSD does not calculate PSNR, SSIM, VMAF, or other decoded-image metrics.
- Narrower format breadth: FFmpeg and MediaInfo support more formats, unusual variants, and codec-container combinations.
- AVI AAC and AVI E-AC-3 mappings are not enabled. Container-only color metadata is not used as a fallback when codec signaling is absent.
- Bitrate differences across unlike resolution, frame rate, or duration are factual, not a quality verdict.

14. Verification and Regression Matrix

14.1 Existing validated VP9 assets

Asset	Statistics
10-second Matroska sample	300 samples; 325 internal frames; 25 superframe samples; 2 KEY; 323 INTER; Q 31-179; zero rejected
Full Matroska file	18,977 samples; 19,176 internal frames; 199 superframe samples; 135 KEY; 19,041 INTER; Q 0-179; zero rejected

14.2 MP4 mapping verification

- Static mapping check: vp09/vp08 -> V_VP9 -> VP9 -> Vp9Frame -> ParseVp9Payload.
- Optional vpcC lookup must remain parent bounded.
- Canonical sample count must equal the resolved MP4 sample-table count.
- Packet/frame accounting, shown/sample consistency, Base Q completeness, and VP9 validation must pass on a real VP9-in-MP4 asset.
- Until that runtime asset is supplied, MP4 VP9 shall be described as implemented or mapped, not regression validated.

14.3 Required invariants

- $\text{InternalFrames} = \text{Parsed} + \text{ShowExisting}$.
- $\text{Parsed} = \text{KeyFrames} + \text{InterFrames}$ for newly coded frames.
- $\text{Histogram count} = \text{Parsed}$ when $\text{Rejected} = 0$.
- $\text{Displayed I} + \text{P} + \text{B} + \text{Other} = \text{container video sample count}$.

15. Maintenance Guidelines

15.1 MP4 VP9 extension rule

- Keep the mapping inside the existing stsd video sample-entry branch.
- Do not introduce a separate MP4-specific VP9 frame parser.
- Retain optional vpcC bytes without requiring them for payload traversal.
- Use the existing sample tables, canonical bounded samples, diagnostics, summary block, and histogram.
- Add a deterministic VP9-in-MP4 regression asset before changing MP4 support status to validated.

15.2 Code that should remain stable

- Canonical model contracts and sample boundary validation.
- Existing Matroska VP9 regression behavior.
- AVI/OpenDML chunk resolution and MP4 parent-bound checks.
- A/B role semantics and snapshot immutability.
- N/A reporting and rejected-unit visibility.

15.3 Future candidates

- Fragmented MP4 support.
- Dedicated VP9-in-MP4 and VP9 Profiles 1, 2, and 3 regression assets.
- Container-color fallback through MP4 colr/nclx and Matroska Colour.
- Additional video codecs such as MPEG-2 Video.
- Optional batch analysis and textual A/B compatibility summary.

Appendix A. Key Types and Responsibilities

Type / state	Responsibility
NativeMediaInfo / NativeTrack	Container-native metadata and codec configuration
LsdMediaModel / LsdTrackModel / LsdSampleModel	Container-neutral tracks and bounded samples
NativeMediaProbe / NativeMp4Probe / NativeAviProbe	Container detection, metadata, and sample indexing
NativeMatroskaPacketScan	Shared canonical video analysis job; historical type name
Vp9BitReader	Bounded MSB-first VP9 bit access
Vp9FrameParser	Superframe split, frame header traversal, reference-slot state, Base Q extraction
Vp9FrameInfo / Vp9ChunkInfo	Per-frame syntax result and per-sample internal-frame collection
CanonicalAudioAnalyzer	MP3, Dolby, and PCM validation/accounting
H264SliceHeaderParser / HvcHvccContextProbe	AVC/HEVC syntax and quantizer extraction
Av1CodecConfigurationProbe	AV1 configuration and sequence data
ReferenceResult	Frozen B file, bitrate bins, QP mode/counts, and summary
PowerShell coordinator	Jobs, state, reports, graphs, comparison, and GUI

Appendix B. Terminology

Term	Definition
Canonical sample	Container-neutral record describing a bounded payload range and timing
Internal VP9 frame	One coded VP9 frame extracted directly from a sample or from its superframe payload
VP9 superframe	One sample containing multiple concatenated coded frames and a trailing size index
Shown frame event	A frame presentation caused by a newly coded shown frame or show_existing_frame
Hidden coded frame	A newly coded frame with show_frame equal to zero
show_existing_frame	Display an existing reference slot without coding a new frame or Base Q Index
KEY_FRAME	VP9 intra frame with key-frame semantics and full reference refresh
INTER_FRAME	VP9 inter-coded frame; hidden alternate-reference frames are included
Base Q Index	Frame-level native quantizer lookup index; 0 through 255 for AV1 and VP9
Current A	Most recently analyzed active result, rendered blue
Reference B	Stored comparison result, rendered gold
Complete	All required relevant units parsed without rejection

Appendix C. Release Identification

File name	LSD-3.0-VP9-MP4.ps1
Version	3.0 - VP9 MP4 mapping build
SHA-256	bf7955bd859a2f37768c981d96d6ab39e23285d7ce334bae7ce3b98d894e62ff
Runtime model	Single PowerShell file with embedded C#
Runtime dependencies	Windows PowerShell 5.1 and .NET Framework classes
External multimedia tools	None
Temporary media files	None
Validated VP9 route	Matroska/WebM V_VP9, Profile 0, 8-bit YUV 4:2:0
Implemented VP9 route	Non-fragmented MP4/M4V/MOV vp09 and vp08; runtime validation pending

Release status. This build retains the validated Matroska/WebM VP9 parser and adds ISO Base Media sample-entry mapping from vp09/vp08 to the same canonical VP9 analysis route. The optional vpcC box is captured as codec private data.

Static integration check. The completed source contains the full route vp09/vp08 -> V_VP9 -> CodecFamily VP9 -> PayloadFormat Vp9Frame -> ParseVp9Payload.

Validation note. No VP9-in-MP4 sample was supplied with this revision. The new mapping therefore requires runtime confirmation before it is labeled regression validated.